DOD-STD-1312-113
26 March 1984

SUPERSEDING
DOD-STD-1312-13
19 October 1983

MILITARY STANDARD

FASTENER TEST METHODS, METRIC

METHOD 113

DOUBLE SHEAR TEST

FSC 53GP

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Fastener Test Methods, Metric, Method 113, Double Shear Test

DOD-STD-1312-113

1. This Military Standard is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specification and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

This standard sets forth standard test procedures for determining compressive loads and shear strengths of fasteners.

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1. SCOPE

1.1 Applicability. This test method covers the procedure and apparatus for testing fasteners in double shear in a half-hole shear fixture.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-45662 Calibration Systems Requirements

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E4 Standard Methods of Verification of Testing Machines

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Testing machine. The testing machine shall be capable of applying a compressive load at a controllable rate. The calibrating system for the machine shall conform to MIL-STD-45662. Its accuracy shall be verified every 6 months by a method complying with ASTM E4, using a calibration device which shall have been calibrated by

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the National Bureau of Standards not more than 2 years prior to its use. The loads of the fasteners tested shall be within the range of the testing machine as defined in ASTM E4.

4.1.2 Test fixture. Test fixtures shall be in accordance with figure 1.

4.1.3 Grip length. Unless otherwise specified, the minimum grip length of the test specimen shall be 3D.

5. DETAIL REQUIREMENTS

5.1 Test procedures. The ultimate double shear strength shall be obtained as follows:

a. The fastener is placed in the receiving half-holes of the fixture base with neither threads nor fillet in bearing.

b. The blade (guillotine) portion is placed in position on the specimen and the preload applied.

c. The test load is then applied at a uniform rate as specified in table I. Load rates for larger or smaller size fasteners shall be calculated to give an initial stress rate of 700 Newtons per minute per square millimeter of nominal double shear area. A tolerance of ±10 percent shall apply on the load rate. Nominal double shear area is twice the nominal shank area for load rate calculation only. The test may be discontinued and the specimen removed without a complete shear failure after the ultimate load has been reached.

TABLE I. Double shear load rates.

NOMINAL DIAMETER	LOAD RATE KILONEWTON/MIN	NOMINAL DIAMETER	LOAD RATE KILONEWTON/MIN
3	10	16	280
4	18	18	350
5	27	20	440
6	40	22	530
7	54	24	630
8	70	27	800
		30	990
10	110	33	1200
12	180	36	1400
14	215	39	1700

The testing laboratory, at their option, may use a constant strain rate which will produce the specified load rate (+10 percent) in the elastic range. That is, the strain rate shall be equal to the initial stress rate of 700 newtons per minute per square millimeter of shear area divided by the elastic modulus in shear.

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6. NOTES

6.1 Test reports. Unless otherwise specified, the following information shall be reported in the product specification for the material being tested:

a. Fastener description.

b. Part number.

c. Material.

d. Manufacturer.

e. Measured diameter of each specimen.

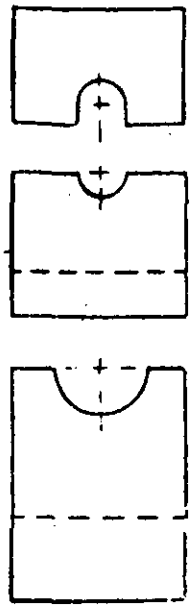
f. Individual ultimate load.

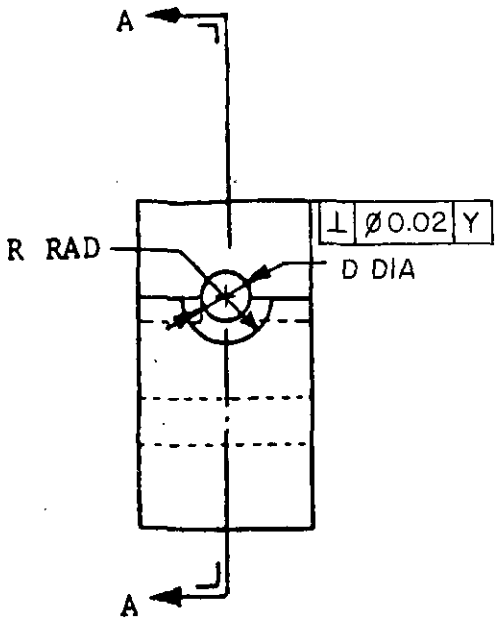
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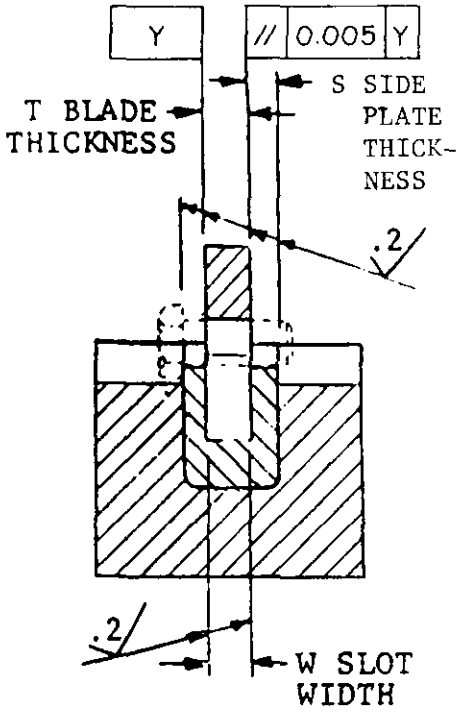
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SECTION A-A

NOMINAL DIA	D		S .025 .000	T +.000 -.025	W +.025 -.000	R
	MAX	MIN				
3	3.025	3.010	1.500	3.000	3.025	3.5
4	4.025	4.010	2.000	4.000	4.025	4.5
5	5.025	5.010	2.500	5.000	5.025	5.
6	6.025	6.010	3.000	6.000	6.025	7.
7	7.025	7.010	3.500	7.000	7.025	8.
8	8.025	8.010	4.000	8.000	8.025	9.
10	10.025	10.010	5.000	10.000	10.025	11.
12	12.025	12.010	6.000	12.000	12.025	13.
14	14.025	14.010	7.000	14.000	14.025	15.
16	16.025	16.010	8.000	16.000	16.025	17.
18	18.025	18.010	9.000	18.000	18.025	18.
20	20.025	20.010	10.000	20.000	20.025	20.
22	22.025	22.010	11.000	22.000	22.025	22.
24	24.025	24.010	12.000	24.000	24.025	25.
27	27.025	27.010	13.500	27.000	27.025	28.
33	33.025	33.010	16.500	33.000	33.025	34.
36	36.025	36.010	18.000	36.000	36.025	37.
39	39.025	39.010	19.500	39.000	39.025	40.

NOTES:
1. Break shear edges, 0.12 minimum. Fixtures shall be reworked when wear results in chamfer or radius of 0.250.
2. For fixture sizes other than those listed, the geometric proportions specified herein shall be maintained.

FIGURE 1. Test fixture.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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